Hzrs 2020 Pimontert
SD204, SN 1248
OSU, SN 0369
Universidad de Concepcion

CALIBRATION CERTIFICATE

NAME	:	RINKO Ⅲ
MODEL	:	ARO-CAV-SA-R
SERIAL No.	:	0380
Parameter	:	Temperature Dissolved Oxygen

JFE

JFE Advantech Co., Ltd.

Temperature Calibration Certificate

Model	:	ARO-CAV-SA-R
Serial No.	:	0380
Date	:	June 28, 2019
Location	:	Production Section
Method	:	Calibration equation is determined from third order regression of samples of the reference temperature against instrument voltages. Samples are taken at approximately 3, 10, 17, 24, and 31 °C.

1. Equation

Instrument temperature[°C] = $A+B \times V+C \times V^2+D \times V^3$ V: Instrument voltage[V]

2. Coefficients

A = −1.247913e+01
B = +2.117776e+01
C = −3.483237e+00
D = +6.543883e−01

3. Calibration results

Reference temperature [°C]	Instrument voltage [V]	Instrument temperature [°C]	Residual error [°C]	Acceptance [°C]	OK/NG
3.403	0.84979	3.404	0.001	±0.020	OK
10.415	1.28766	10.412	−0.003	±0.020	OK
17.359	1.74609	17.363	0.004	±0.020	OK
24.207	2.19891	24.204	−0.003	±0.020	OK
31.082	2.63324	31.083	0.001	±0.020	OK

4. Verification

Criteria of judgement : Residual error of the instrument temperature at arbitrary point is within the acceptance value.

Reference temperature [°C]	Instrument temperature [°C]	Residual error [°C]	Acceptance [°C]	Judgement
20.287	20.298	0.011	±0.020	Passed

Examined K. Shimotan

Approved M. Ujinaki

JFE Advantech Co., Ltd.

Dissolved Oxygen Calibration Certificate

Model	:	ARO-CAV-SA-R
Serial No.	:	0380
Date	:	July 01, 2019
Location	:	Production Section
Method	:	Calibration is performed with the nitrogen gas (zero) and the oxygen saturated water (span) kept by air bubbling.
Film No.	:	191625BA

1. Equation

DO[%] =G+H × P'

Here, P'[%] consists of the coefficients A-F determined by the initial calibration.

2. Coefficients

A =	-4.392753e+01	E =	+4.000000e-03
B =	+1.423079e+02	F =	+5.620000e-05
C =	-2.808449e-01	G =	+0.000000e+00
D =	+1.044900e-02	H =	+1.000000e+00

3. Verification

Criteria of judgement : Residual error of the instrument DO at arbitrary point is within the acceptance value. The test is performed 3 times.

Acceptance: ±0.5% of full scale

Test for DO 0 %

	Test condition		Instrument DO [%]	Residual error [%]	Acceptance [%]	Judgement
	Atm. pressure [hPa]	Reference DO [%]				
1st	1006.4	0.00	-0.02	-0.02	±1.00	Passed
2nd	1006.4	0.00	-0.03	-0.03	±1.00	Passed
3rd	1006.4	0.00	-0.03	-0.03	±1.00	Passed

Test for DO 100 %

	Test condition			Instrument DO [%]	Residual error [%]	Acceptance [%]	Judgement
	Water T. [°C]	Atm. pressure [hPa]	Reference DO [%]				
1st	25.1	1006.1	99.27	99.05	-0.22	±1.00	Passed
2nd	25.0	1006.1	99.27	99.04	-0.23	±1.00	Passed
3rd	25.0	1006.1	99.27	99.31	0.04	±1.00	Passed

Examined M. TAKEISHI

Approved M. Ujinaki

JFE Advantech Co., Ltd.